

Exercises sheet 1: Counting

Exercise 1. Two consecutive experiments are conducted. The first experiment has m possible outcomes. For the i^{th} outcome of the first experiment, the second experiment has n_i possible outcomes, where $i = 1, \dots, m$. What is the total number of possible outcomes for both experiments combined?

Exercise 2. How many possible initials (consisting of 2 letters) can a person have? How many students are needed in a class to guarantee that at least two of them share the same initials?

Exercise 3. The 10 volumes of an encyclopedia are arranged randomly. In how many ways can they be arranged if there are no restrictions? How many ways are there if volumes 7, 6, and 5 must be placed next to each other in that specific order?

Exercise 4. How many 3-letter words can be formed with no restrictions? How many can be formed with at least 2 consonants? With exactly 2 consonants and 1 vowel? With at most 1 vowel?

Exercise 5. On Eid, you plan to visit 7 family members. In how many ways can you visit all of them if all visits are made on the same day? How many ways are there if you visit 4 of them on Eid and the remaining 3 the next day?

Exercise 6. How many football teams can be formed from 22 people without restrictions? How many teams can be formed if there are 3 goalkeepers and one of them must be included?

Exercise 7. A company consists of 2 managers, 3 administrators and 11 workers. Four people are invited to an exhibition. How many ways can they be chosen if i.) There are no restrictions? ii.) There are 1 manager, 2 administrators, and 1 worker? iii.) Workers and administrators can not be together?

Exercise 8. A class of 100 students is to be divided into five groups of 15, 18, 20, 21 and 26 students respectively. How many choices do we have for this division?

Exercise 9. In a committee of 10 people, a chairperson, a secretary and a treasurer should be selected

1. Knowing that accumulation is not allowed, in how many ways can these offices be allocated if
 - i. There are no restrictions? ii. A and B can not be together? iii. C and D are together or not at all?
 - iv. F must have a task? v. F only accepts the task of president?
2. Knowing that accumulation is allowed, in how many ways can these offices be allocated if there are no restrictions?

Exercise 10. In the exam, the student has to answer 5 out of 8 questions. How many choices does he have? How many choices would he have if he had to answer at least 3 out of the first 5 questions?

Exercise 11. How many different words can be formed from the letters of the words: ALGER, ANNABA, TIZI-OUZOU, MISSISSIPPI.

Exercise 12. *In how many ways can 8 people be seated*

1. *In a row, if: i.) there are no restrictions? ii.) two persons A and B stay together? iii.) there are 4 adults and 4 children, and each adult has only children as neighbors and vice versa? iv.) the number of adults is 6, they must stay together, v.) there are 4 children, each accompanied by his mother who stays next to him?*
2. *Around a table, redo i.) and iii.)?*

Exercise 13. *Four Algerians, three Libyans and two Tunisians must be lined up in a race. Athletes of the same nationality must stay together. How many arrangements can we imagine?*

Exercise 14. *m boys and n girls are to be arranged, where $m, n \in \mathbb{N}$. Find the number of ways this can be done in each of the following cases.*

I) In a row:

1.) There are no restrictions; 2.) No boys are adjacent ($m < n + 1$); 3.) The n girls form a single block; 4.) A particular boy and a particular girl must be adjacent.

II) Around a table redo 1.) and 3.)

Exercise 15. *In a car park, there are 30 cars in a row, 8 of them have broken down. How many possibilities are there for parking 30 vehicles (always in a row), if every pair of broken down vehicles is separated by at least two good vehicles? Then generalize?*

Exercise 16. *An elevator leaves the ground floor with 8 athletes (5 Algerians and 3 Tunisians). When it reaches the 6th floor, it is empty. In how many ways can we perceive the departure of the 8 athletes if: no distinction is made? If we distinguish them by their nationality?*

Exercise 17. *One city in Algeria has 5 clinics. Food poisoning affected 15 people. In how many ways can these poisoned people be spread over the 5 clinics, (The poisoned people are not discernible) if 1) the number of poisoned people received by each clinic is not limited? 2) a certain clinic of the five receives exactly 7 poisoned people? 3) Two particular clinics A and B receive at least 3 people?*

Exercise 18. *How many pieces are there in a game of dominoes, knowing that on each piece there are two symbols chosen among: blank, 1, 2, 3, 4, 5, 6? blank, 1, 2, ..., 9? Generalize to triominoes and tetraominoes.*

Exercise 19. *Consider r distinguishable flags (numbered from 1 to r) and n poles numbered from 1 to n (large enough to carry the flags). How many ways can the flags be arranged on the poles? start with $r = n = 2$ (draw a picture). What difference would it make if we put each flag in a box (the boxes being indistinguishable) and put the r flags in n drawers?*

Exercise 20. *Consider $E = \{1, 2, \dots, p\}$ and $F = \{1, 2, \dots, q\}$. Determine the number of*

- 1. Applications from E in F ?*
- 2. Injections from E in F ?*
- 3. Bijection from E in F ?*

Exercise 21. Five prizes (one per each course) are to be awarded to deserving student selected from a class of 25. How many results do we have?

Exercise 22. The postman arrives with 5 letters in a building where there are 7 mailboxes. Determine the number of ways to place the 5 letters in the 7 mailboxes if: i. a mailbox can contain all 5 letters. ii. a mailbox can only contain one letter.

Exercise 23. A group of 5 people carries out two independent draws to designate two people in the group to perform certain tasks, without specifying the order of the tasks. How many possible combinations are there?

Exercise 24. Consider a population of N individuals. We take n successive samples from this population, with replacement. How many sequences of lengths n exists in which individual A is taken: 1. k times ? 2. k times in the first r draws ? 3. for k^{th} times in the r^{st} draws ?

Exercise 25. 1. Expand $(2x^3 - 3y^2)^5$ and $(x^2 - 2y + 4)^3$.

2. Give the coefficients of the monomials $x^{14}y^5z^3$ and xy^2z^3 in the expansion of $(2x^2 - 3y + z)^{15}$.

Exercise 26. How many solutions does the equation $x_1 + x_2 + \cdots + x_r = n$ have, in $\mathbb{N} \cup \{0\}$, for which k are exactly zeros?

Exercise 27. Give the number of solutions in $\mathbb{N} \cup \{0\}$ for:

1. $x + y + z = 10$ where $x \geq 2, y \geq 1, z \geq 2$.
2. $x + y + z + t = 10$ where $x \geq 2, y \geq 1, t \geq 2$.
3. $x_1 + x_2 + \cdots + x_r \leq n$.
4. $x_1 + x_2 + \cdots + x_r \leq n$ where $x_1 \geq 1, x_2 \geq 2, \dots, x_r \geq r$.

Exercise 28. Find the number of integer solutions to the equation $x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 60$, in each of the following cases:

(i) $x_i \geq i - 1$ for each $i = 1, 2, \dots, 6$; (ii) $x_1 \geq 2, x_2 \geq 5, 2 \leq x_3 \leq 7, x_4 \geq 1, x_5 \geq 3$ and $x_6 \geq 2$.

Exercise 29. Find the number of integer solutions to the equation $x_1 + x_2 + x_3 + x_4 = 30$

in each of the following cases (i) $x_i \geq 0$ for each $i = 1, 2, 3, 4$; (ii) $2 \leq x_1 \leq 7$ and $x_i \geq 0$ for each $i = 2, 3, 4$; (iii) $x_1 \geq -5, x_2 \geq -1, x_3 \geq 1$ and $x_4 \geq 2$. (iv.) x_i is positive and odd for each $i = 1, 2, 3, 4$.

Exercise 30. Find the number of quadruples (w, x, y, z) of nonnegative integers which satisfy the inequality $w + x + y + z \leq 1992$.

Exercise 31. Prove, first using combinatorial argument and then analytically, that:

$$1. \binom{n+m}{r} = \sum_{k,l/k+l=r} \binom{n}{k} \binom{m}{l} = \sum_{k=0}^r \binom{m}{k} \binom{n}{r-k} \text{ Vandermonde's formula. Deduce that}$$

$$i. \sum_{k=0}^n \binom{n}{k}^2 = \binom{2n}{n} \quad ii. \sum_{k=0}^n k \binom{n}{k}^2 = \frac{n}{2} \binom{2n}{n} \quad iii. \sum_{k=0}^{2n} (-1)^k \binom{2n}{k}^2 = (-1)^n \binom{2n}{n}.$$

$$2. \sum_{m_1, \dots, m_q / \sum_1^q m_i = m} \binom{n_1}{m_1} \binom{n_2}{m_2} \dots \binom{n_q}{m_q} = \binom{n_1 + n_2 + \dots + n_q}{m}.$$

$$3. \binom{n}{n_1, n_2, \dots, n_r} = \binom{n}{n_1} \binom{n - n_1}{n_2} \dots \binom{n - n_1 - \dots - n_{r-1}}{n_r}.$$

$$4. \binom{n}{n_1, n_2, \dots, n_r} = \binom{n-1}{n_1-1, n_2, \dots, n_r} + \binom{n-1}{n_1, n_2-1, \dots, n_r} + \dots + \binom{n-1}{n_1, n_2, \dots, n_r-1}.$$

Exercise 32. Give the approximate value at 10^{-3} to $(0, 980)^4$, $(1, 001)^{100}$, and $(\frac{899}{900})^{50}$.

Exercise 33. p balls numbered from 1 to p are placed in n indistinguishable boxes ($p \geq n$). Let D_n^p be the number of possible arrangements such that no box is empty.

$$1. \text{ Show that } D_n^p = nD_n^{p-1} + D_{n-1}^{p-1} \quad p \geq n + 1.$$

$$2. \text{ Calculate } a/D_1^p \text{ and } D_n^n \quad b/D_2^p \text{ and } D_3^p \quad c/D_{p-1}^p, D_{p-2}^p \text{ and } D_{p-3}^p.$$

Exercise 34. Calculate

$$1. S_1 = \binom{n}{0} + \binom{n}{2} + \binom{n}{4} + \dots \quad \text{and} \quad S_2 = \binom{n}{1} + \binom{n}{3} + \binom{n}{5} + \dots \quad 2. \prod_{k=0}^n \binom{n}{k} = \binom{n}{0} \binom{n}{1} \dots \binom{n}{n}$$

$$3. \sum_{k=0}^n \frac{1}{k+1} \binom{n}{k} \quad 4. a/ \sum_{k=1}^n k \binom{n}{k} \quad b/ \sum_{k=1}^n k^2 \binom{n}{k} \quad c/ \sum_{k=1}^n k^3 \binom{n}{k} \quad d/ \sum_{k=1}^n (-1)^k k \binom{n}{k}.$$

Resume a/, b/ and c/ using a combinatorial argument [consider the number of possible subsets containing one element x , 2 elements x and y ($x = y$) and x, y, z respectively.]

$$5. S_1 = \binom{n}{0} + \binom{n}{3} + \binom{n}{6} + \dots, \quad S_2 = \binom{n}{1} + \binom{n}{4} + \binom{n}{7} + \dots, \quad S_3 = \binom{n}{2} + \binom{n}{5} + \binom{n}{8} + \dots$$

Exercise 35. Establish

$$1. \binom{n}{k} \binom{k}{r} = \binom{n-r}{k-r} \binom{n}{r} \quad 2. \sum_{k=r}^n \binom{n}{k} \binom{k}{r} = \binom{n}{r} 2^{n-r}.$$

$$3. \sum_{k=0}^n k \binom{n}{k} = n 2^{n-1}. \quad 4. \sum_{k=r}^n \binom{n}{k} \binom{k}{r} (-1)^{n-k} = 0.$$

$$5. \binom{n}{k} = \binom{n-2}{k-2} + 2 \binom{n-2}{k-1} + \binom{n-2}{k} \quad 6. \binom{n}{k} = \binom{n-3}{k-3} + 3 \binom{n-3}{k-2} + 3 \binom{n-3}{k-1} + \binom{n-3}{k},$$

Generalize the first two previous formulas.

Exercise 36. Show that: 1. $\binom{n}{k}^2 > \binom{n-1}{k} \binom{n+1}{k} > \binom{n}{k-1} \binom{n}{k+1}$. 2. $\binom{n+1}{k+1} = \sum_{j=0}^{n-k} \binom{n-j}{k}$.

Deduce that $\sum_{k=1}^n k$, $\sum_{k=1}^n k(k+1)$, $\sum_{k=1}^n k(k+1)(k+2)$, $\sum_{k=1}^n k^2$, $\sum_{k=1}^n k^3$.

Exercise 37. Show that

$$1. \quad \binom{n}{r} = \sum_{k=0}^r \binom{n-1}{k} = \sum_{k=0}^r \binom{n-1}{r-k}. \quad 2. \quad \binom{n+m}{r} = \sum_{k=0}^r \binom{n}{k} \binom{m}{r-k}.$$

Exercise 38.

1. Let f and g be two function of class C^∞ . Establish the Leibnitz's formula $(f \cdot g)^{(n)} = \sum_{k=0}^n \binom{n}{k} f^{(k)} g^{(n-k)}$.
2. Show that for all $x \in]-1, 1[$, $\frac{1}{(1-x)^{p+1}} = \sum_{n=p}^{\infty} \binom{n}{p} x^{n-p}$.

Exercise 39 (Pascal's inversion formula).

Let $(f_n)_{n \geq 0}$ and $(g_n)_{n \geq 0}$ be two sequences of real numbers such that, for any integer $n \geq 0$,

$$f_n = \sum_{k=0}^n \binom{n}{k} g_k = g_0 + \binom{n}{1} g_1 + \binom{n}{2} g_2 + \cdots + \binom{n}{n} g_n.$$

Show that, for any integer $n \geq 0$,

$$g_n = \sum_{k=0}^n (-1)^{n-k} \binom{n}{k} f_k = g_0 + (-1)^n \binom{n}{1} f_1 + (-1)^{n-1} \binom{n}{2} f_2 + \cdots + \binom{n}{n} f_n.$$

1. **Surjections:** Let $E = \{1, 2, \dots, p\}$ and $F = \{1, 2, \dots, q\}$, note that $S_{p,q}$ the number of surjections from E in F .
 - a) What is the number of applications from E in F , whose image is I_k , $\text{Card} I_k = k$, for any I_k fixed part ($k \leq q$)?
 - b) What is the number of applications from E in F , whose image has the cardinal k ?
 - c) By classifying the applications from E in F , show the relation: $q^p = \sum_{k=1}^q \binom{q}{k} S_{p,k}$.
 - d) Deduce that $S_{p,k} = \sum_{q=k}^p (-1)^{q-k} \binom{q}{k} k^p$ with the convention $S_{p,0} = 0$ for all $p \in \mathbb{N}$.
2. **Derangements:** the number of permutations of the set $\{1, \dots, n\}$ having exactly k fixed points, denoted by P_n^k . A permutation without fixed point is called a derangement and it is denoted by $D_n (= P_n^0)$.
 - a. Show that $P_n^k = \binom{n}{k} D_{n-k}$, with the convention $D_0 = 1$.
 - b. Show that, we have for all $n \geq 0$, $\mathcal{P}_n = \sum_{k=0}^n P_n^k$.
 - c. Establish that, for all integer $0 \leq k \leq n$: $P_n^k = \frac{n!}{k!} \sum_{p=0}^{n-k} \frac{(-1)^p}{p}$. Deduce the expression of D_n .

Supplementary Exercises

Exercise 40. Find the number of ways of forming a group of $2k$ people from n couples, where $k, n \in \mathbb{N}$ with $2k \leq n$, in each of the following cases:

1. There are k couples in such a group;
2. No couples are included in such a group;
3. At least one couple is included in such a group;
4. Exactly two couples are included in such a group.

Exercise 41. There are n sticks lined up in a row, and k of them are to be chosen.

1. How many choices are there?
2. How many choices are there if no two of the chosen sticks can be consecutive?
3. How many choices are there if there must be at least l sticks between each pair of chosen sticks?

Exercise 42. At a particular company, employees have a seven digit security code. Five of the digits must come from $\{1, 4, 5, 7\}$ with repetition allowed. The remaining digits must come from $\{2, 3, 9\}$ with no repetition.

1. How many security codes are possible?
2. How many of these begin with 1?
3. How many of these end with 5?
4. How many of these neither begin with 1 nor end with 5?
5. Suppose that this company has 150,000 employees. Should they be that two employees have the same security code?

Exercise 43. I) Twenty different books are to be put on five book shelves, each of which holds at least twenty books.

1. How many different arrangements are there if you only care about the number of books on the shelves (and not which book is where)?
2. How many different arrangements are there if you care about which books are where, but the order of the books on the shelves doesn't matter?
3. How many different arrangements are there if the order on the shelves does matter?

- II) 1. Let n and k be positive integers. Give a combinatorial proof that $\sum_{k=1}^n k \binom{n}{k}^2 = n \binom{2n-1}{n-1}$.
2. What is the coefficient of $x^{30}y^{20}z^{50}$ in the expansion of $(x - 2y + 3z - \frac{1}{2}w)^{100}$?

Exercise 44. 1. Prove in two different ways the following identity $k \binom{n}{k} = n \binom{n-1}{k-1}$.

2. Calculate $\sum_{r=0}^n 3^r \binom{n}{r}$, and then give a combinatorial proof for the derived identity.